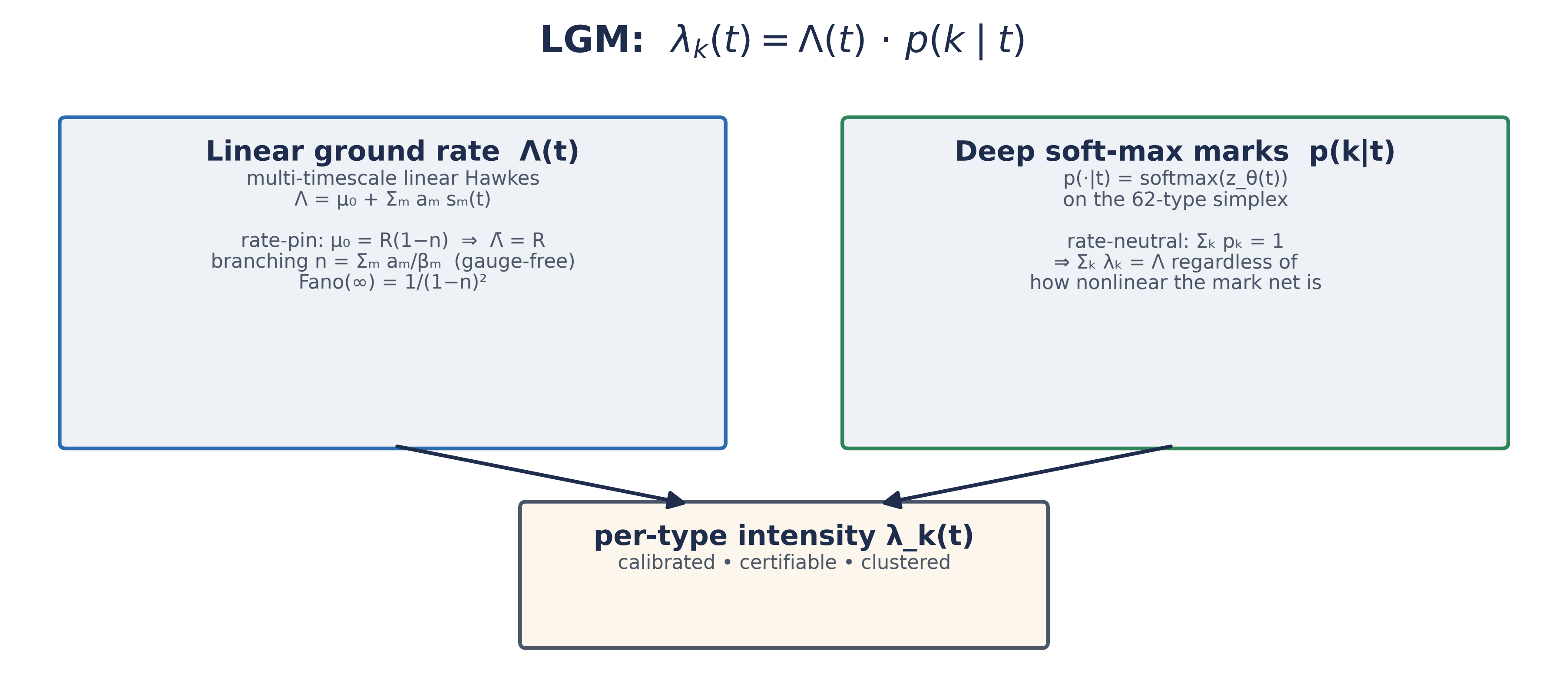
LGM: $\lambda_k(t) = \Lambda(t) \cdot p(k \mid t)$
Linear ground rate Λ(t)
multi-timescale linear Hawkes
$\Lambda = \mu_0 + \Sigma_m a_m s_m(t)$
rate-pin: $\mu_0 = R(1-n) \Rightarrow \bar{\Lambda} = R$
branching $n = \Sigma_m a_m/\beta_m$ (gauge-free)
$\mathrm{Fano}(\infty) = 1/(1-n)^2$
Deep soft-max marks p(k|t)
p(·|t) = softmax(z_θ(t))
on the 62-type simplex
rate-neutral: $\Sigma_k p_k = 1$
$\Rightarrow \Sigma_k \lambda_k = \Lambda$ regardless of
how nonlinear the mark net is
per-type intensity λ_k(t)
calibrated • certifiable • clustered